Structural Design Report

N2 / G+1 Storey

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Abstract

G+1 story low rise RCC structure is *Designed & Drafted* using *BNBC2020*. Various supporting simulation and drafting tools are used.

1 Project Information

- **Project:** G+1 story in Narshingdi
- **Location:** Narshingdi
- **Building Use:** Residential
- **Structural System:** RCC frame

2 Design Basis

2.1 Codes & Standards

- BNBC 2020 (Loads, Concrete/Steel)
- ASCE, ACI (Design Load Combination)

2.2 Properties

Table 1: Material Properties

Item	Symbol	Value
Concrete strength	f'_c	2500 psi
Steel yield	f_y	725 000 psi
Unit weight (RCC)	γ_c	150 lb/ft^3
Brick Unit Weight	γ_b	120 lb/ft^3

3 Loads & Load Combinations

3.1 Gravity Loads

Loads that are used as gravity load are.

Table 2: Gravity load on zones

Zone	Load
1st Floor Slab	SW, LL_42, LL_RW, DL_FF, DL_FPW
1st Floor Veranda	SW, LL_60, DL_FF
Stair	SW, LL_100, DL_FF
Roof	SW, LL_63, DL_FF_40, DL_FPW_Ppt
OHWT	SW, LL_OHWT, DL_FPW

Table 3: Loads

Item	Symbol	Value
Load	SW	calc
Load	FF	calc or 25 psf
Load	RW	calc or 50 psf
Load	LL_{res}	42 psf
Load	$LL_{res,ver}$	60 psf
Load	$LL_{res,roof}$	63 psf
Load	LL_{OHWT}	calculated 312.5 psf
Load	FPW_{int}	calculated 437.5 plf
Load	FPW_{roof}	calculated 150 plf
Load	FPW_{OHWT}	calculated 500 plf

3.2 Lateral Loads

- Seismic: Only in detailing.
- Wind: Only in detailing.

3.3 Load Combinations

$$1.2 * DeadLoad + 1.6 * LiveLoad$$

4 Structural Model

Structural 3D model is shown in Figure 1.

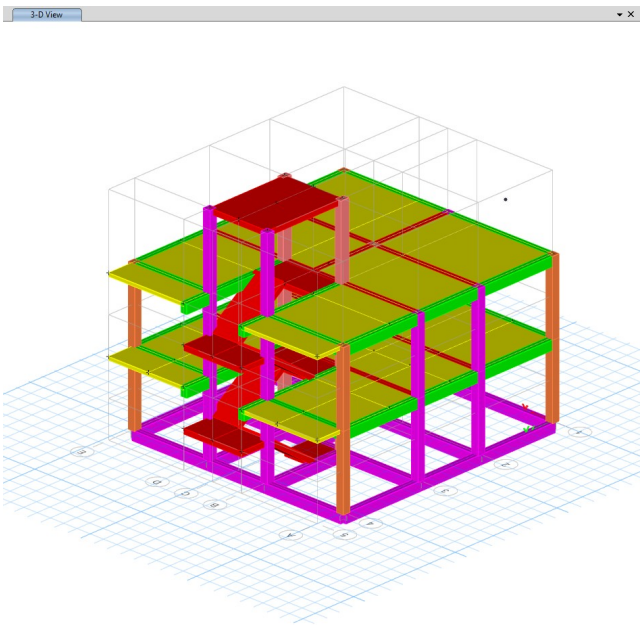

Figure 1: 3d view

5 Structural Analysis and Design

5.1 Slab

Figure 2 shows 1 directional rebar and thickness adequacy of GF level slabs. Similarly other slabs are adequate in both direction.

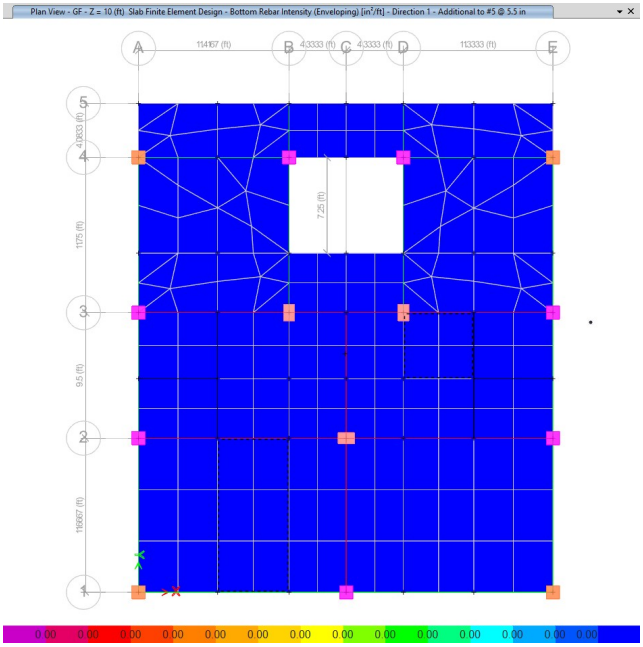

Figure 2: Slab thickness and rebar adequacy check.

5.1.1 Beam

Rebar area required square inch for GF level beams are showen in Figure 3. All the beams are designed and detailed accordingly.

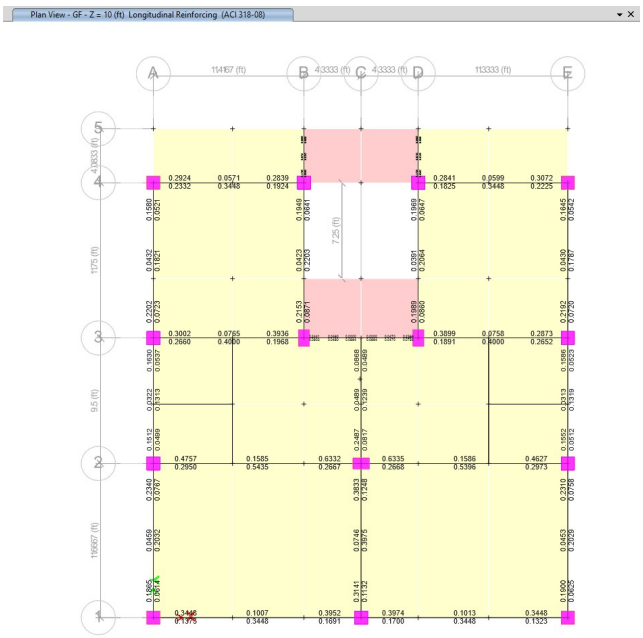

Figure 3: Rebar requirements for GF level Beams.

5.1.2 Beam

Rebar area required in percentage for one elevation view for columns are showen in Figure 4. All the columns required the minimum rebar area.

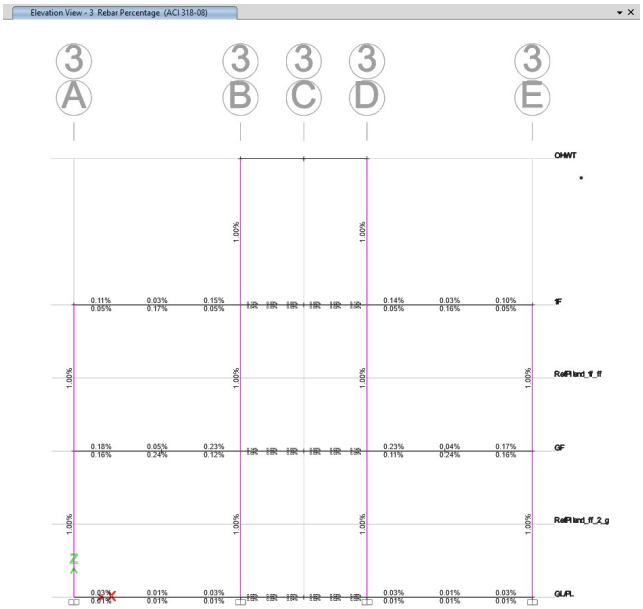

Figure 4: Rebar percentage requirements column at Elevation 3

5.1.3 Base reaction

Base reaction for the combined load of Dead and Live are shown in Figure 5. Using 1.1*(DL+LL) the footing size of columns are calculated and Using 1.2DL + 1.6LL load combination the footing thickness and their corresponding rebar detailing are done.

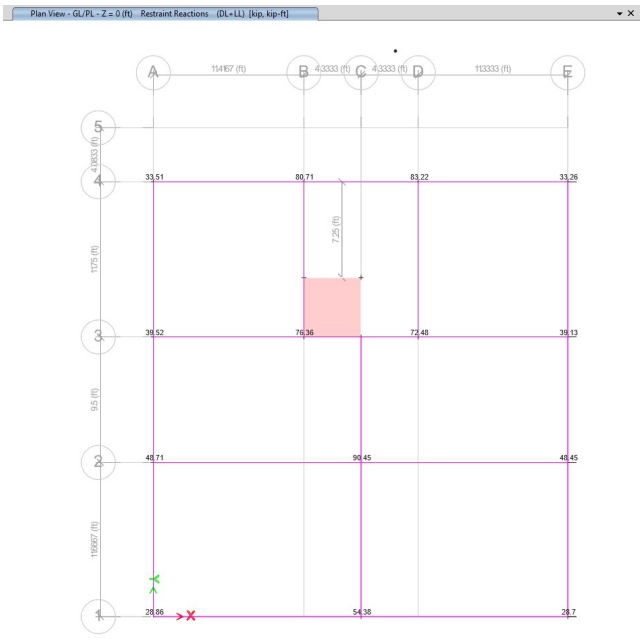

Figure 5: Base reactions considering Dead + Live load